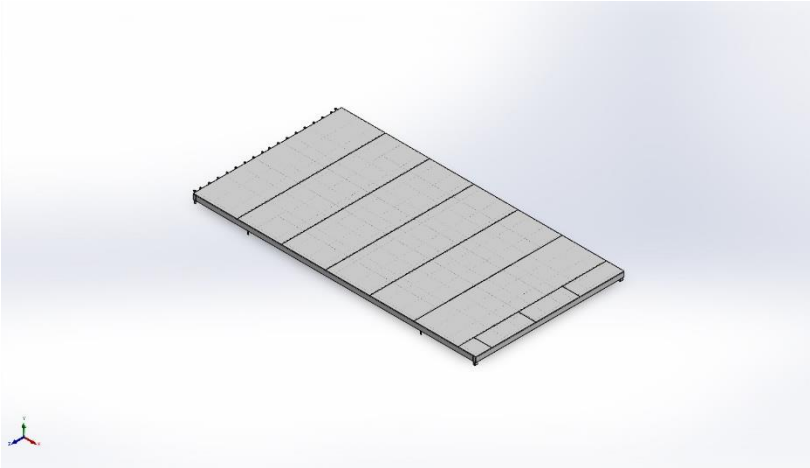
NIHVA Technologies Pvt. Ltd.
Plot No. 187,1st floor, G Block, above Kotak Bank,
MIDC, Thermax Chowk, Pimpri-Chinchwad, Maharashtra 411019



Simulation of 70010309

Study name: Scissor Platform Analysis
Analysis type: Static

Table of Contents

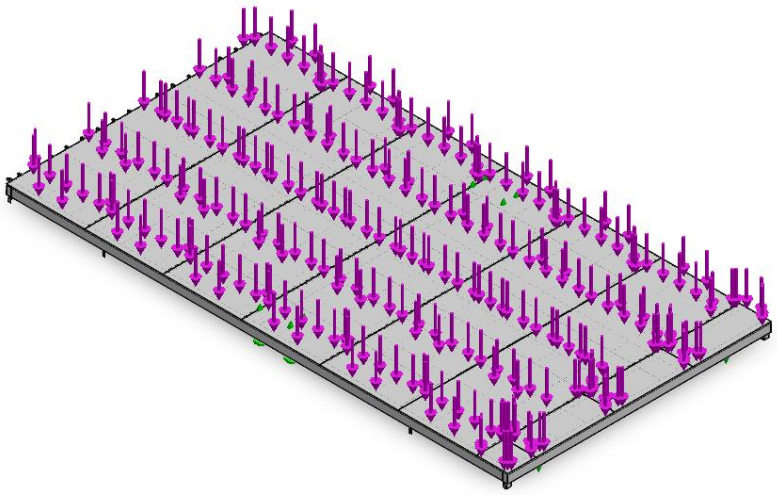
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Description

Scissor platform having a capacity of 30 ton

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Model Information





Model name: 70010309
Current Configuration: Default

Comments:
Platform size-8000L X 4200W
with 10 mm sheet covering

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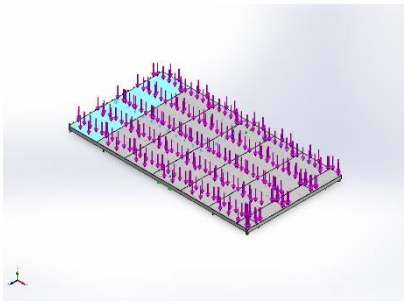
Study Properties

Study name	Static 3 from [Static 1] test
Analysis type	Static
Mesh type	Mixed Mesh
Thermal Effect:	On
Thermal option	Include temperature loads
Zero strain temperature	298 Kelvin
Include fluid pressure effects from SOLIDWORKS Flow Simulation	Off
Solver type	Intel Direct Sparse
Inplane Effect:	Off
Soft Spring:	Off
Inertial Relief:	Off
Incompatible bonding options	Automatic
Large displacement	Off
Compute free body forces	On
Friction	Off
Use Adaptive Method:	Off
Result folder	SOLIDWORKS document (F:\3903 for analysis\PPC-3903_SCISSOR LIFT TABLE_CAPACITY-30 TON_8000X4000\TOP PLATFORM)

Units

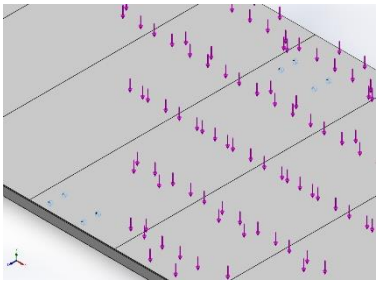
Unit system:	SI (MKS)
Length/Displacement	mm
Temperature	Kelvin
Angular velocity	Rad/sec
Pressure/Stress	N/m^2

Material Properties

Model Reference	Properties		Components
	**Name:**	M.S	All components of structural platform assembly (std .material)
	Model type:	Linear Elastic Isotropic	
	Default failure criterion:	Unknown	
	Yield strength:	2.20594e+08 N/m^2	
	Tensile strength:	3.99826e+08 N/m^2	
	Elastic modulus:	2.1e+11 N/m^2	
	Poisson's ratio:	0.28	
	Mass density:	7,800 kg/m^3	
	Shear modulus:	7.9e+10 N/m^2	
	Thermal expansion coefficient:	1.3e-05 /Kelvin	

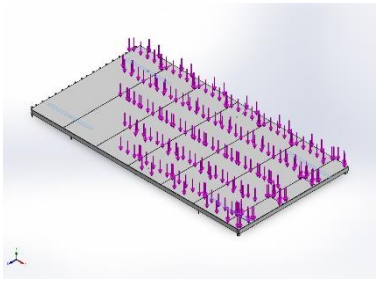
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Loads and Fixtures

Fixture name	Fixture Image	Fixture Details
Fixed-1		**Entities:** 8 face(s) **Type:** Fixed Geometry

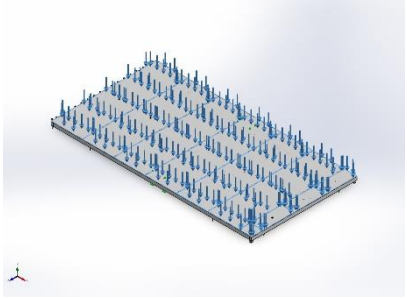
Resultant Forces

Components	X	Y	Z	Resultant
Reaction force(N)	-0.000671387	90,772	0.000656128	90,772
Reaction Moment(N.m)	0	0	0	1e-33

Fixture name	Fixture Image	Fixture Details
Roller/Slider-1		**Entities:** 4 face(s) **Type:** Roller/Slider

Resultant Forces

Components	X	Y	Z	Resultant
Reaction force(N)	0	209,228	0	209,228
Reaction Moment(N.m)	0	0	0	1e-33

Load name	Load Image	Load Details
Force-1		**Entities:** 10 face(s) **Type:** Apply normal force **Value:** 300,000 N

Mesh information

Mesh type	Mixed Mesh
Mesher Used:	Blended curvature-based mesh
Jacobian points for High quality mesh	16 Points
Jacobian check for shell	On
Maximum element size	501.127 mm
Minimum element size	25.0564 mm
Mesh Quality	Draft
Remesh failed parts independently	Off

Mesh information - Details

Total Nodes	153108
Total Elements	430965
Time to complete mesh(hh;mm;ss):	00:35:23
Computer name:	DESIGNPC3

Resultant Forces

Reaction forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	-0.000671387	300,000	0.000656128	300,000

Reaction Moments

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	1e-33

Free body forces

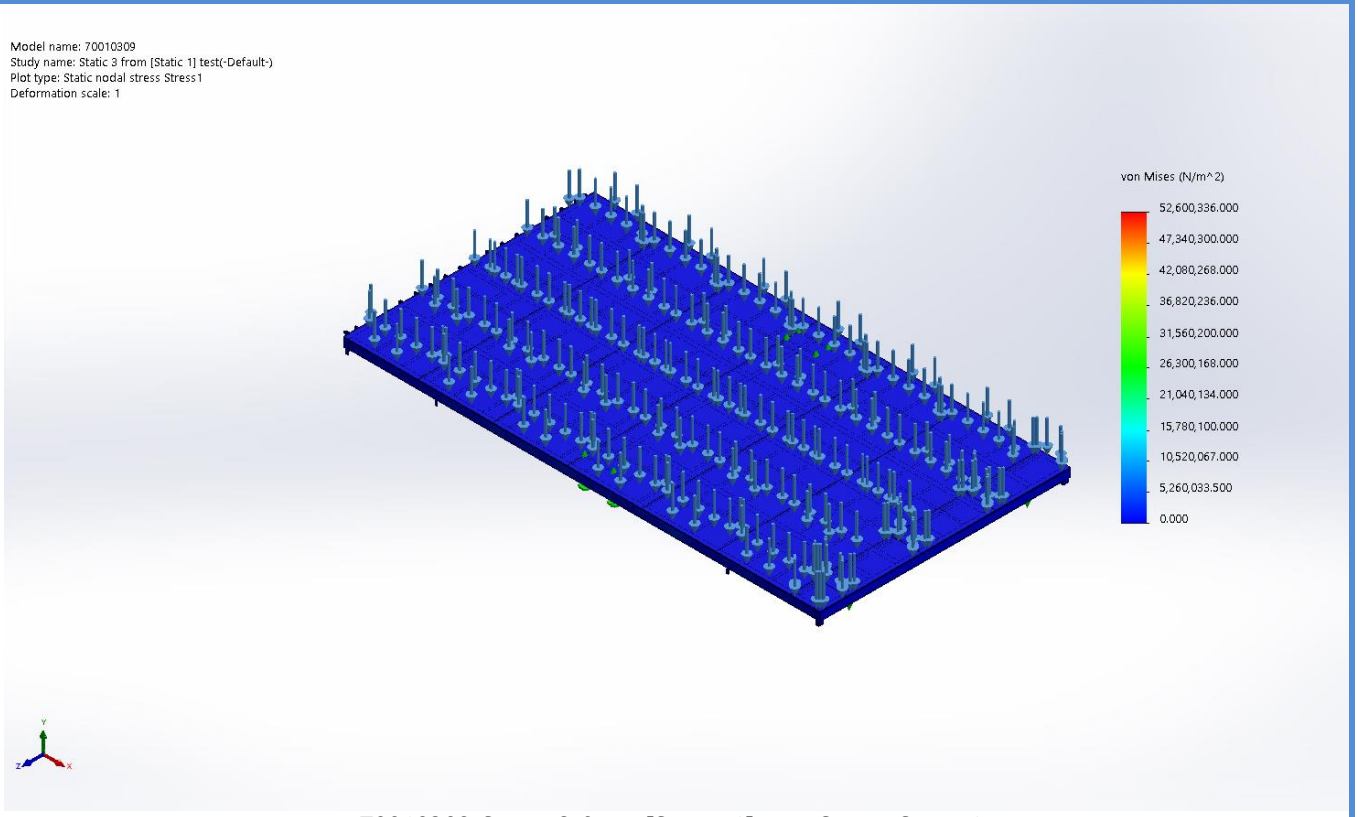
Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	-0.0227385	-0.132835	0.355579	0.380262

Free body moments

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	4.21132e-07	7.00355e-06	1.7077e-06	7.22103e-06

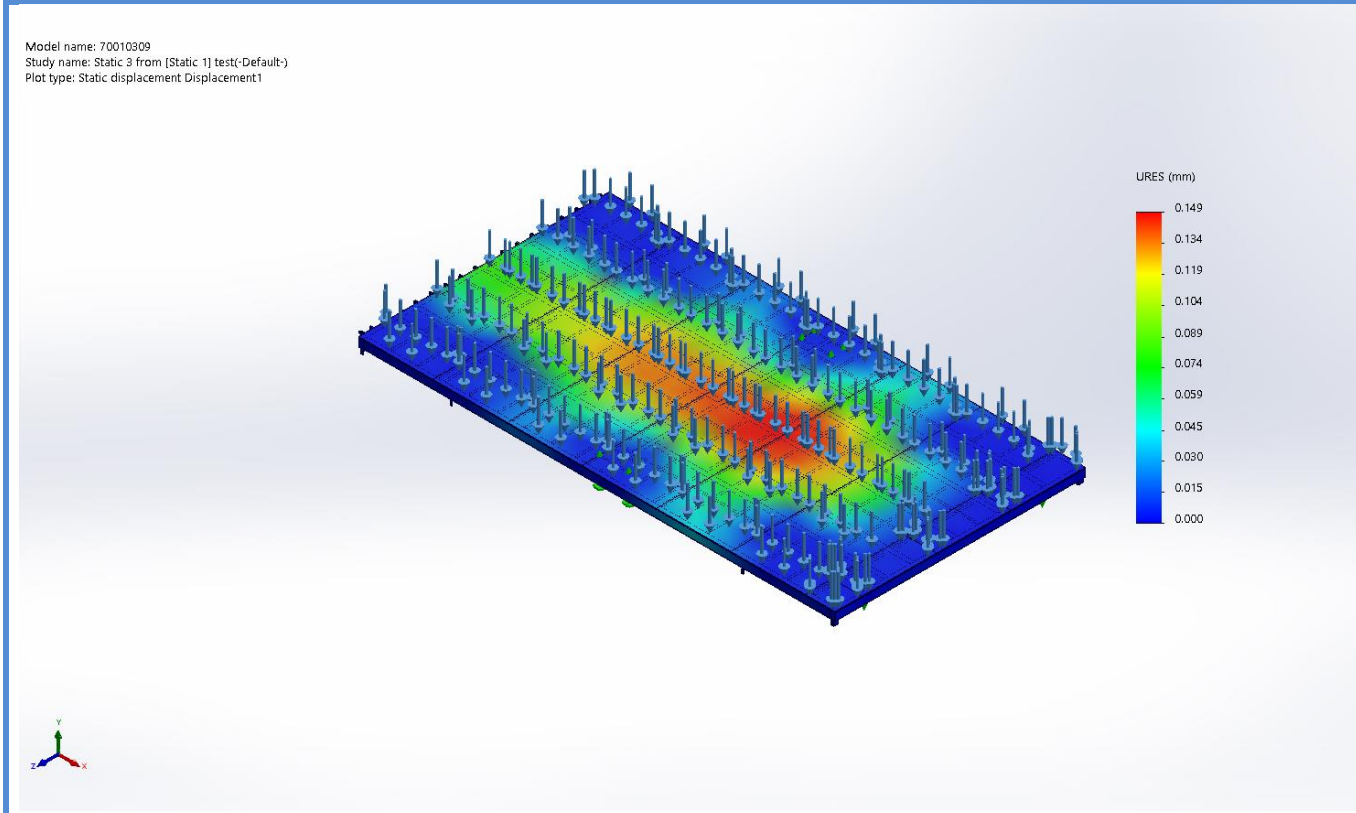
Study Results-

Name	Type	Min	Max
Stress1	VON: von Mises Stress	0.000N/m^2 Node: 152729	52,600,336.000N/m^2 Node: 64344



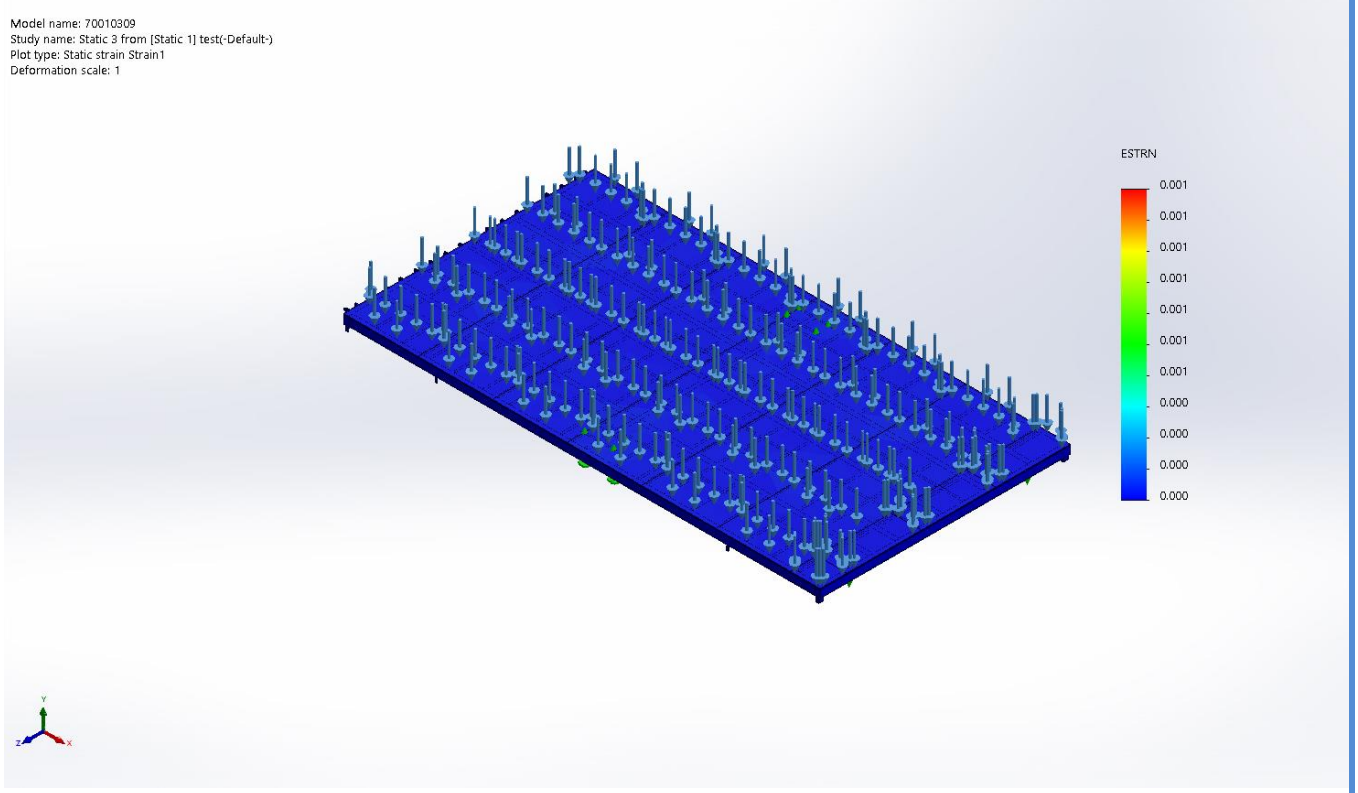
70010309-Static 3 from [Static 1] test-Stress-Stress1

Name	Type	Min	Max
Displacement1	URES: Resultant Displacement	0.000mm Node: 149879	0.149mm Node: 149793



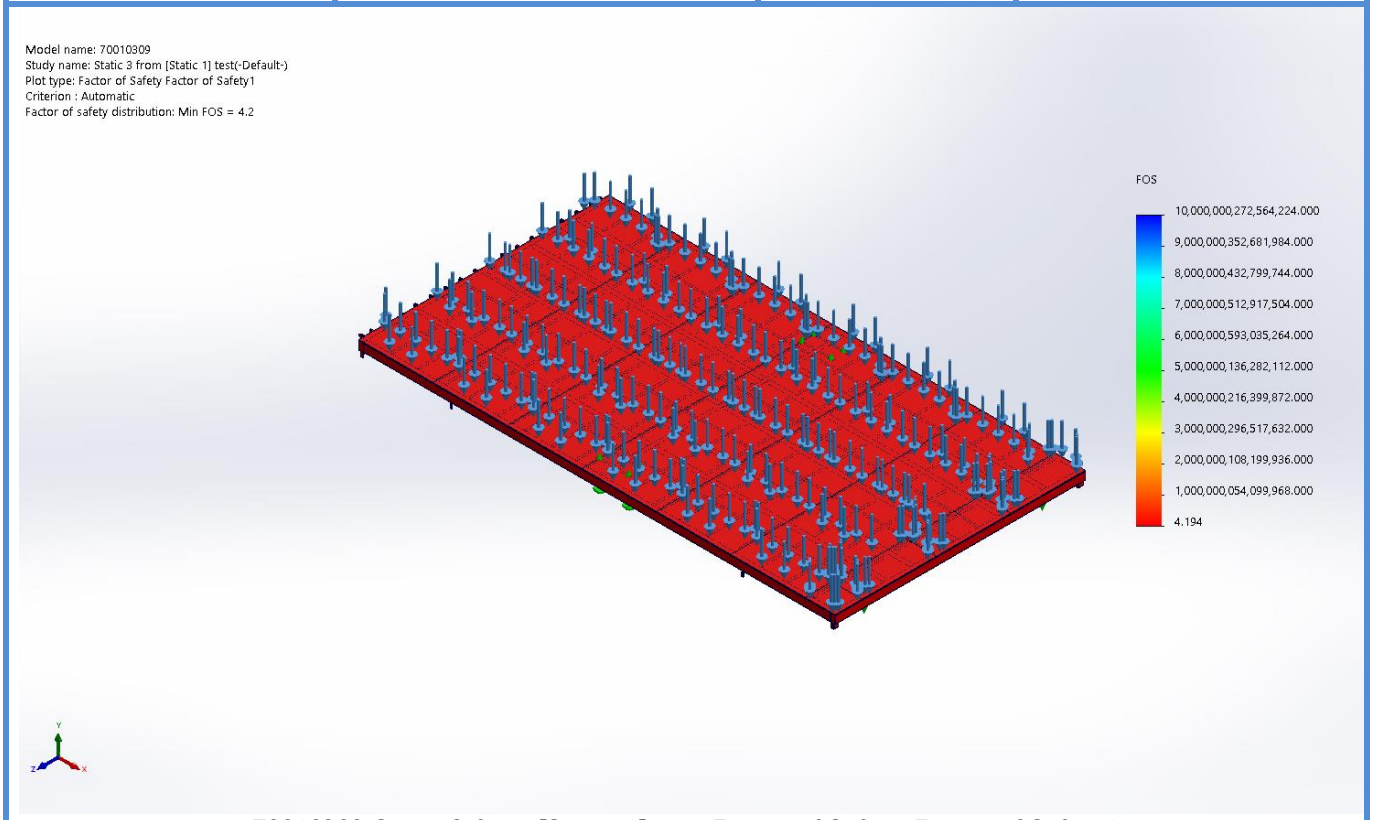
70010309-Static 3 from [Static 1] test-Displacement-Displacement1

Name	Type	Min	Max
Strain1	ESTRN: Equivalent Strain	0.000 Element: 430121	0.001 Element: 187810



70010309-Static 3 from [Static 1] test-Strain-Strain1

Name	Type	Min	Max
Factor of Safety1	Automatic	4.194 Node: 64344	10,000,000,272,564,224.000 Node: 152729



70010309-Static 3 from [Static 1] test-Factor of Safety-Factor of Safety1

Conclusion-

All the results of simulation work on platform assembly are shown above –
displacement -0.149 mm
factor of safety-4.19

Therefor platform under 30 ton load is safe